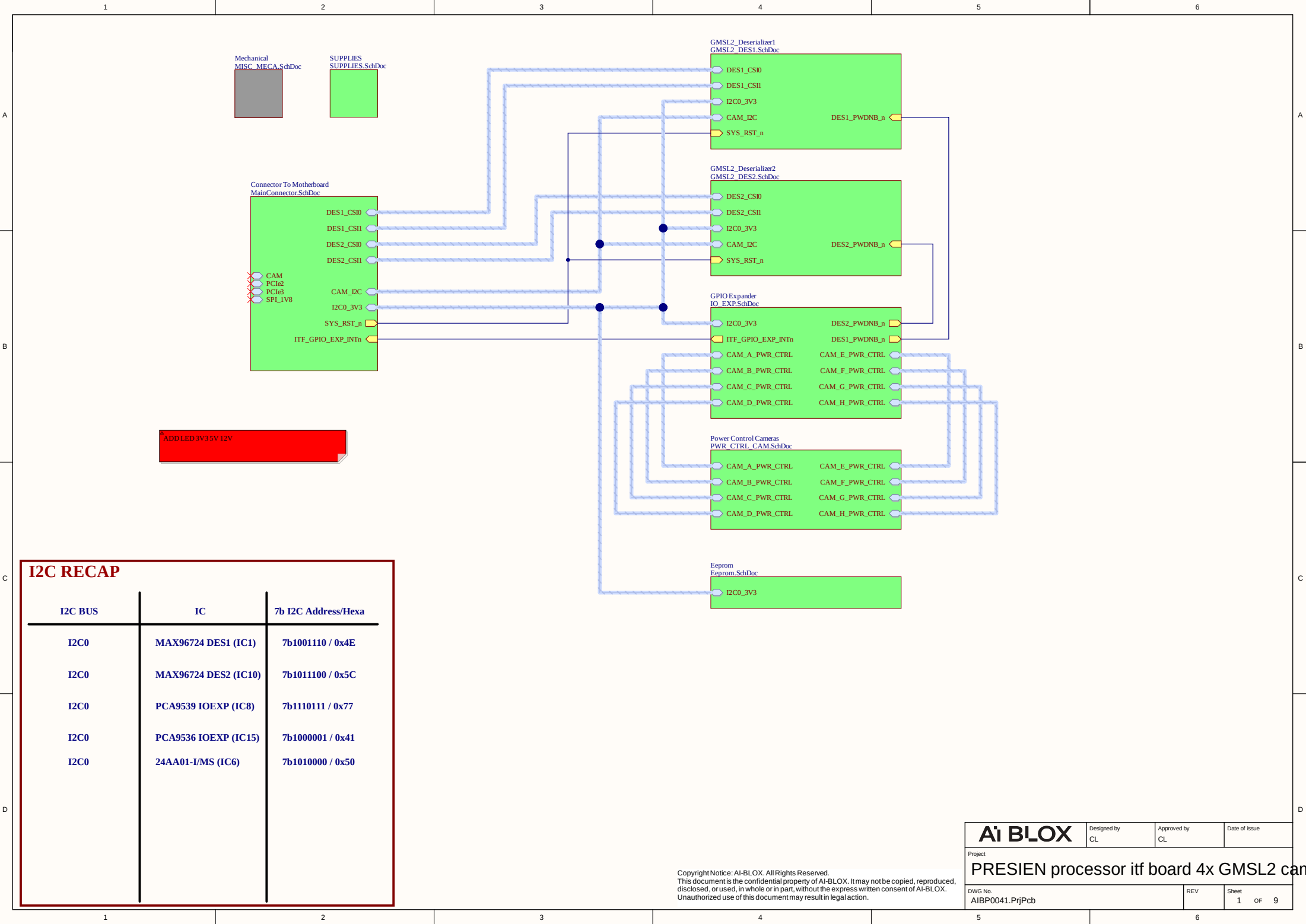
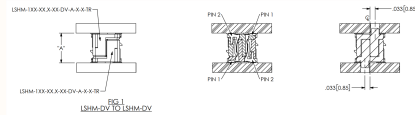




I2C RECAP		
I2C BUS	IC	7b I2C Address/Hexa
I2C0	MAX96724 DES1 (IC1)	7b1001110 / 0x4E
I2C0	MAX96724 DES2 (IC10)	7b1011100 / 0x5C
I2C0	PCA9539 IOEXP (IC8)	7b1110111 / 0x77
I2C0	PCA9536 IOEXP (IC15)	7b1000001 / 0x41
I2C0	24AA01-I/MS (IC6)	7b1010000 / 0x50

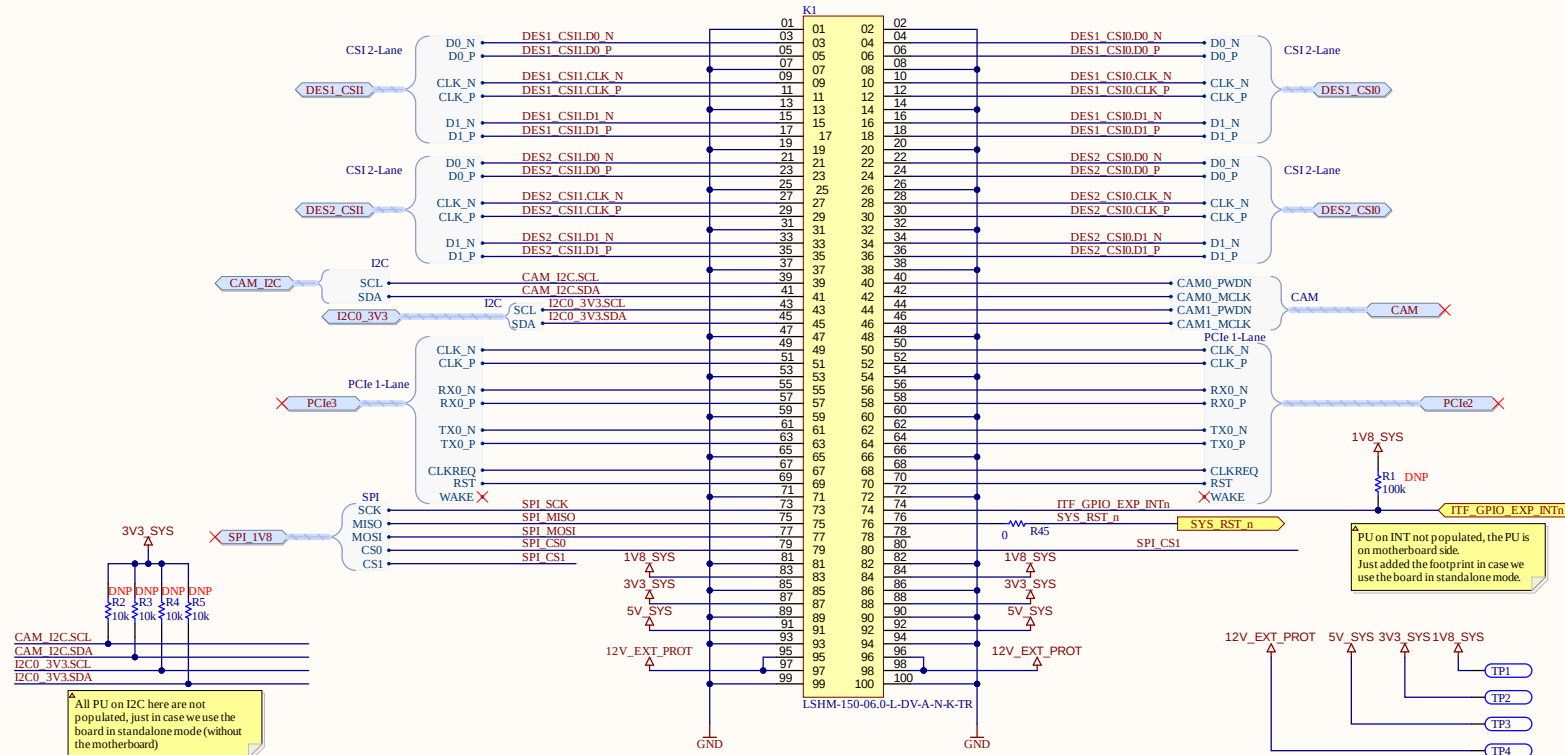
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Pin#1 of the motherboard connector becomes pin#2 of the mezzanine connector. Pinout must be swapped at the other side.

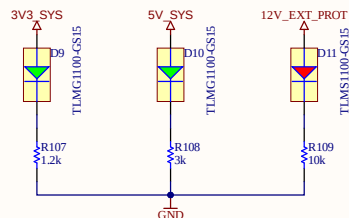


CONNECTOR TO/FROM MOTHERBOARD

DES1_CS10/DES1_CS11 correspond to Jetson CS10/11 respectively
DES2_CS10/DES2_CS11 correspond to Jetson CS12/3 respectively
On motherboard



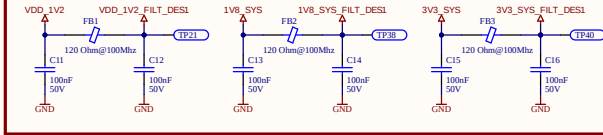
Supplies LEDs



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AIBP0041.PrjPcb		2	9	

PWR SUPPLIES FILTERING



Input Data rate calculation (SIO Side):
Camera is ZED X One CSI Wide. Output format is RAW10 (from camera datastream) which means 10bits/pixel.
Preferred capture mode is 960x500/60fps (from Presien Specification)
It leads to 960x500x10x60 = 345.6Mbps per camera video
= Overhead Protocol

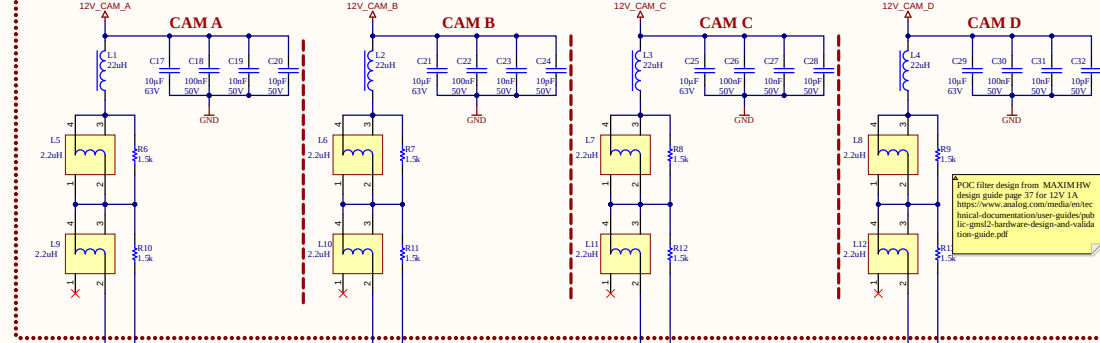
Layout note: CSI Diff Pair should be 100R Diff + All CSI Lane whatever the bus should be length matched all together. See picture below

Table 10-4. MIPI CSI D-PHY Interface Signal Routing Requirements

Parameter	Requirement	Units	Notes
Max Data Rate (per data lane) for High-Speed mode	2.5	Gbps	
Max Frequency (for Low Power mode)	10	MHz	
Number of lanes	1	lane	
Reference plane	GND		
Trace impedance: Diff pair / SE	90 / 100 / 45 / 50	Ω	±10%
Via proximity (signal to reference)	< 0.05 (0.8)	mm (mil)	
Intra-pair trace spacing	0.15mm	mm	Can be adjusted to meet Differential Impedance.
Trace spacing: Microstrip / Stripline	2x / 2x	dielectric	
Max PCB treatment delay	48	ps	
Max insertion loss		dB	
1 Gbps	3.00		
1.5 Gbps	2.90		
2.5 Gbps	1.92		
Max trace delay / length		ps (mm)	
1 Gbps (Stripline/Microstrip)	2526 (4231) / 2487 (4231)		
1.5 Gbps	1913 (3139) / 1885 (3139)		
2.5 Gbps	960 (150) / 886 (150)		
Max intra-pair skew	1	ps	
Max trace delay skew between DQ and CLK	40 / 26.7 / 16	ps	DQ includes all the data lines associated with a single clock. This may be 2 differential data lanes for a 4-lane interface, or 4 differential data lanes for a x8 interface.

Keep critical traces away from other signal traces or unrelated power traces/lines or power supply components

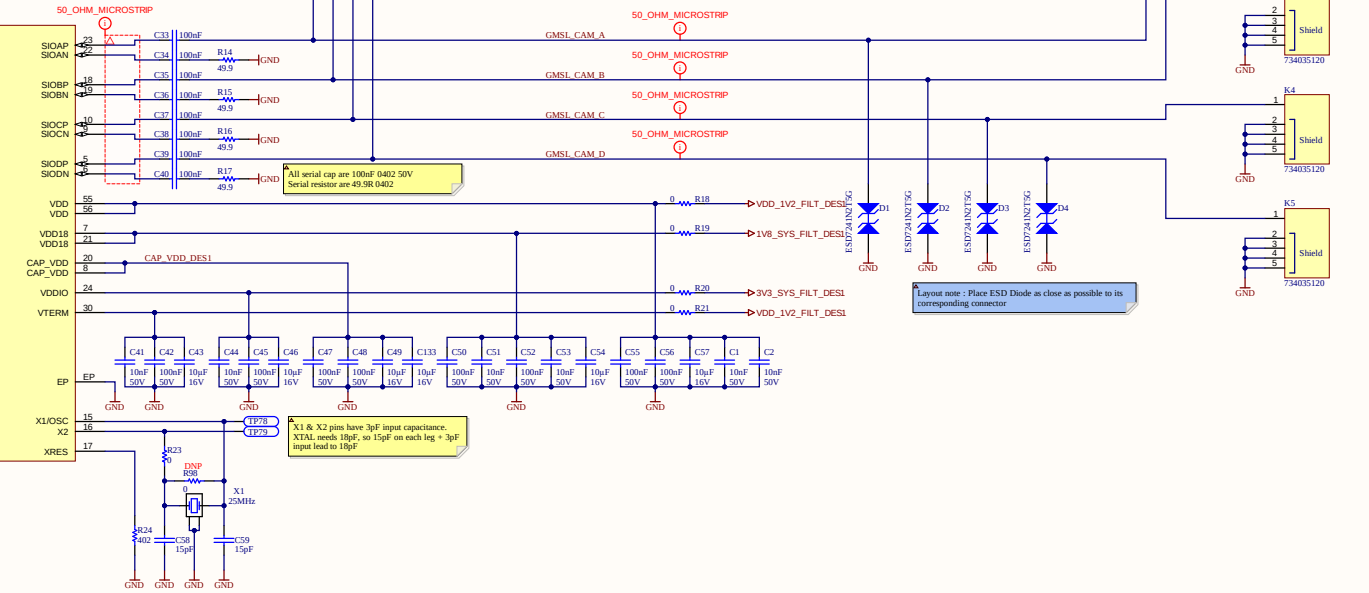
12V INJECTION (PoC)



POC filter design from MAXIMHW design guide page 37 for 12V 1A
<https://www.analog.com/media/en/technical-documentation/user-guides/pub-lic-gmsl2-hardware-design-and-validation-guide.pdf>

Layout note: Place link capacitor as close as possible to its corresponding SIO Pin.

Layout: Remove "copper ring" in internal layer to avoid capacitive effects

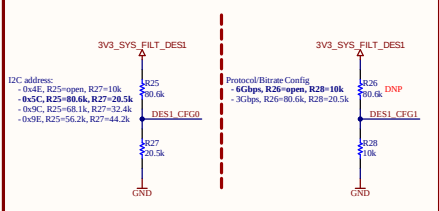


All serial cap are 100nF 0402 50V
Serial resistor are 49.9R 0402

Layout note: Place ESD Diode as close as possible to its corresponding connector

X1 & X2 pins have 3pF input capacitance.
XTAL needs 18pF, so 15pF on each leg + 3pF input load to 18pF

CONFIG PINS



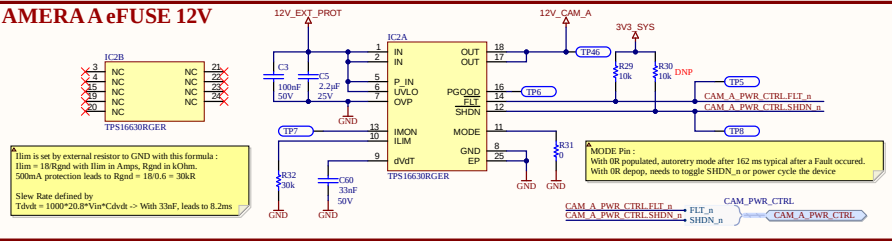
I2C address:
-0x4E, R25=open, R27=10k
-0x5C, R25=80.6k, R27=20.5k
-0x9C, R25=68.1k, R27=32.4k
-0x9E, R25=56.2k, R27=44.2k

Protocol/BiRate Config
-6Gbps, R26=open, R28=10k
-3Gbps, R26=80.6k, R28=20.5k

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Project PRESIEN processor itf board 4x GMSL2 camera			
DWG No. AIBP0041_PriPcb	REV	Sheet 3 of 9	

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CAMERA A eFUSE 12V



Power Consumption of Camera ZED X ONE GS WIDE is :

- Min : 0.7W <==> 59mA @12V

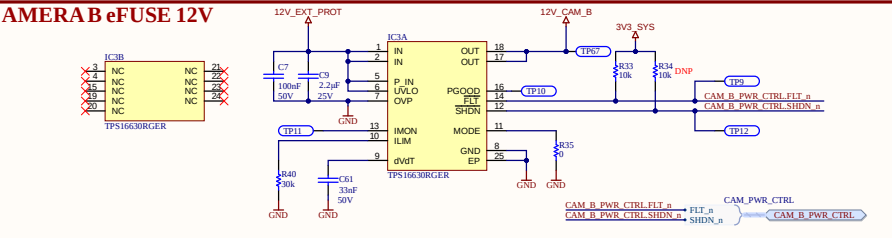
- Max : 0.94W <==> 79mA @12V

For reference :

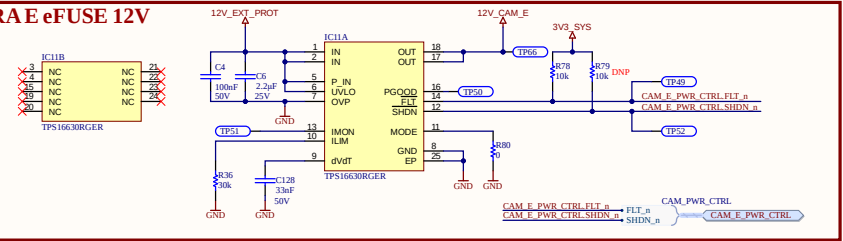
https://www.stereolabs.com/docs/get-started-with-zed-link/power_requirements

For later design change, if we need to reduce the price, use the following eFuse : TPS1641x.

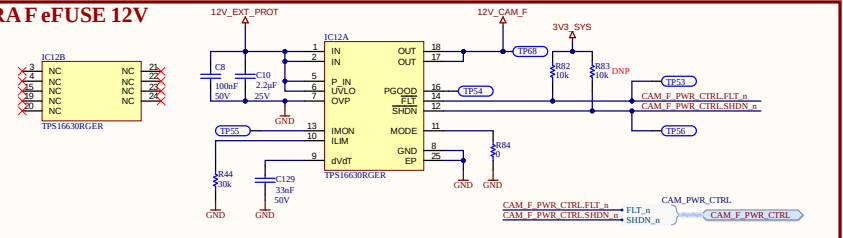
CAMERA B eFUSE 12V



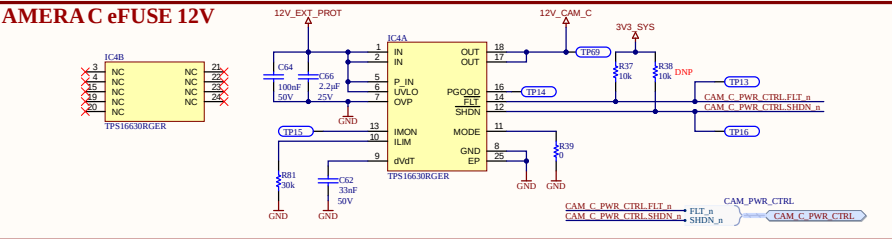
CAMERA E eFUSE 12V



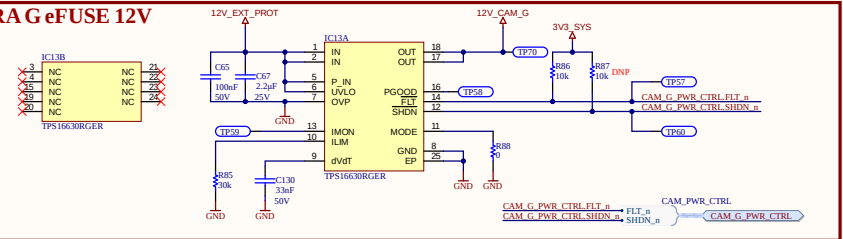
CAMERA F eFUSE 12V



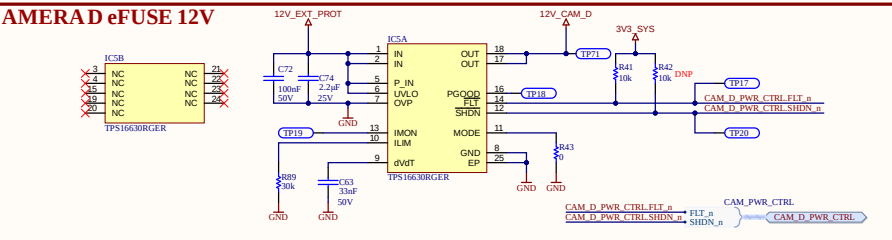
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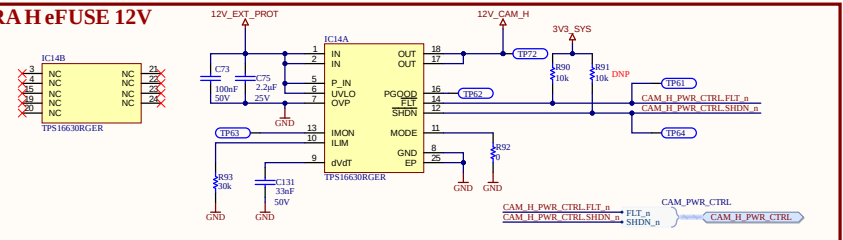
CAMERA G eFUSE 12V



CAMERA D eFUSE 12V



CAMERA H eFUSE 12V



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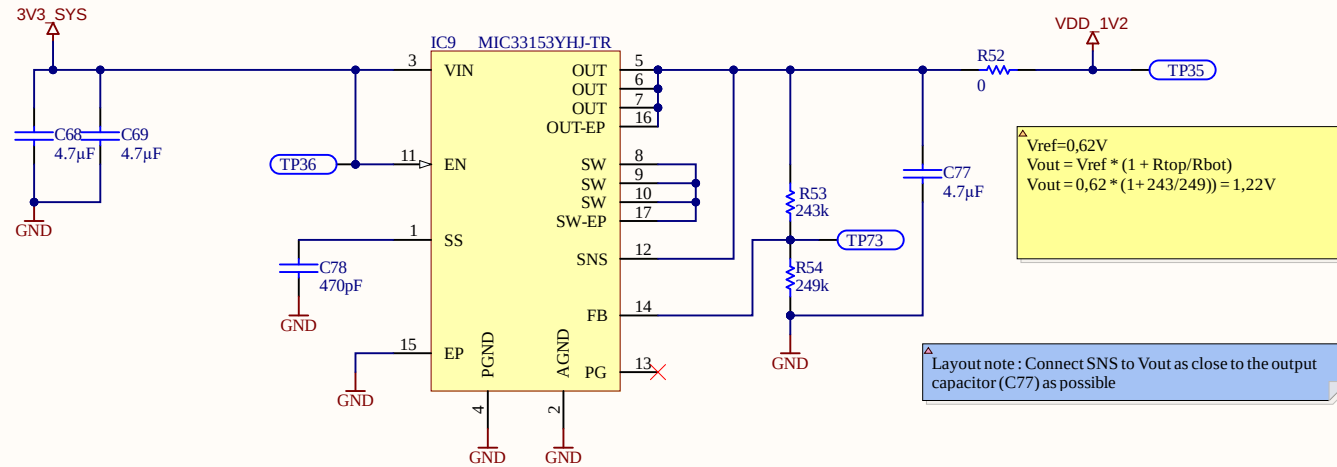
REV

Sheet

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3V3 SYS ==> 1V2 @1.2A max



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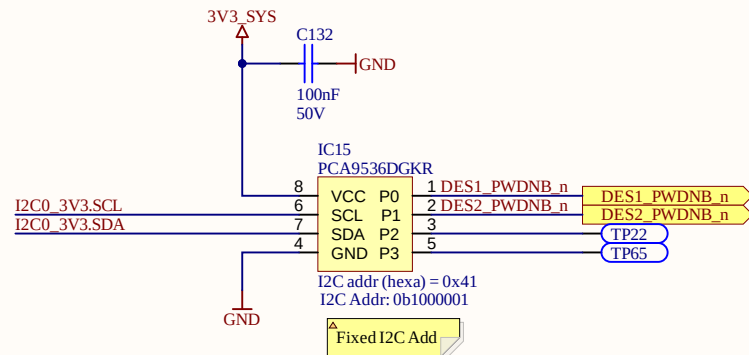
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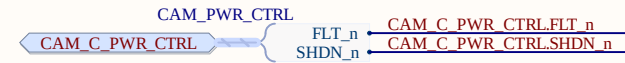
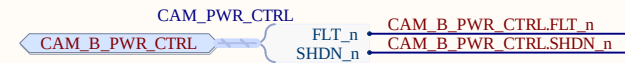
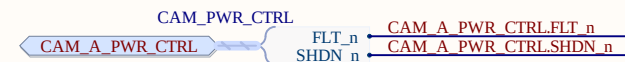
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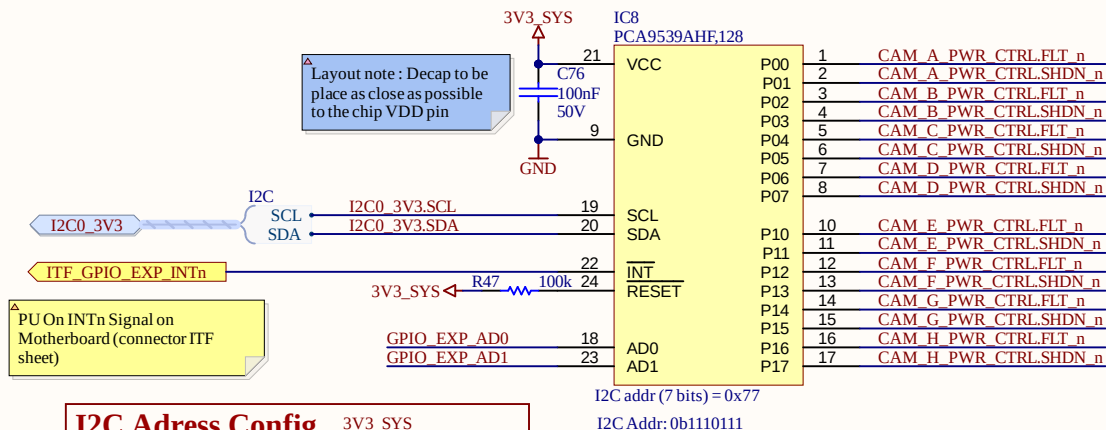
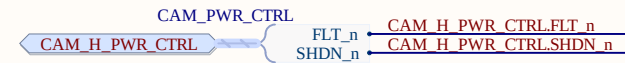
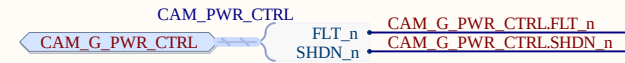
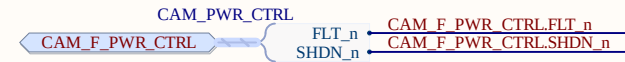
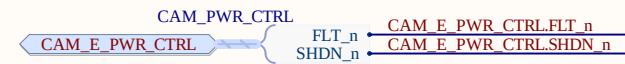


POWER DOWN_n To DES1 & DES2

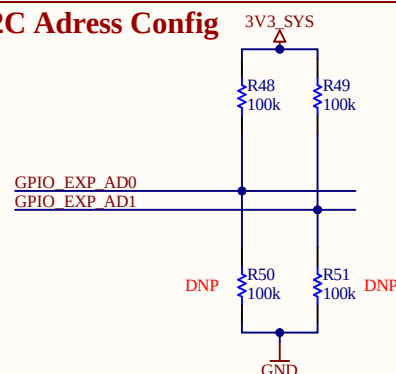
IO FROM eFuse 12V CAMERA A/B/C/D



IO FROM eFuse 12V CAMERA E/F/G/H



I2C Address Config



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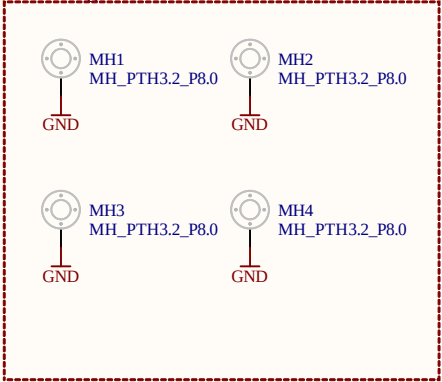
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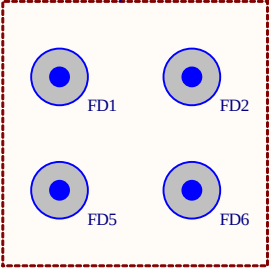
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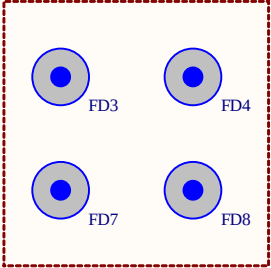
Mounting holes PCB



Fiducials Top



Fiducials Bottom



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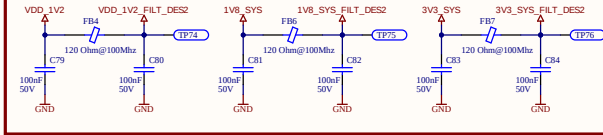
PRESIEN processor itf board 4x GMSL2 camera

DWG No.
AIBP0041.PrjPcb

REV

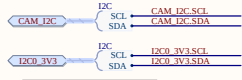
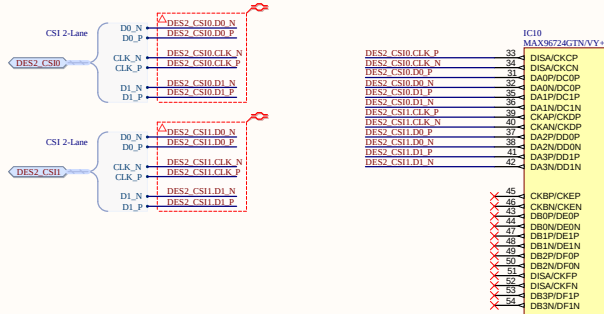
Sheet
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PWR SUPPLIES FILTERING



Input Data rate calculation (SIO Side) :
Camera is ZED X One GS Wide. Output format is RAW10 (from camera dataset) which means 10bits/pixel.
Preferred capture mode is 960x600@60fps (from Presien Specification)
It leads to 960x600x10x60 = 345.6Mbps per camera video
→ Overhead Protocol

Layout note : CSI Diff Pair should be 100R Diff + All CSI Lane whatever the bus should be length matched all together. See picture on GMSL2_DES1 sheet



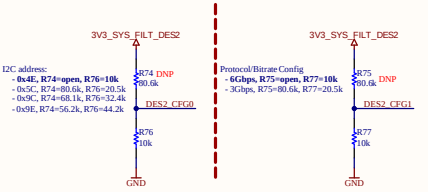
PU On I2C Signals on Motherboard Jetson
Test point I2C on DES1 Sheet



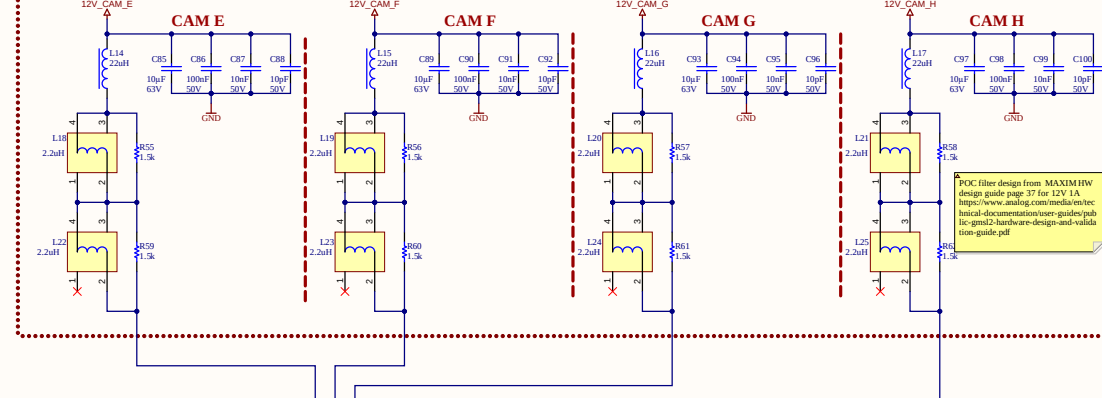
PWDNB is used here as power-down or reset (hold the signal low for at least 1us for reset)

By default, we use the RESET_n from mainboard to reset MAX96724. If Reset via IO/EXP needed, populate R95 and depop R97

CONFIG PINS

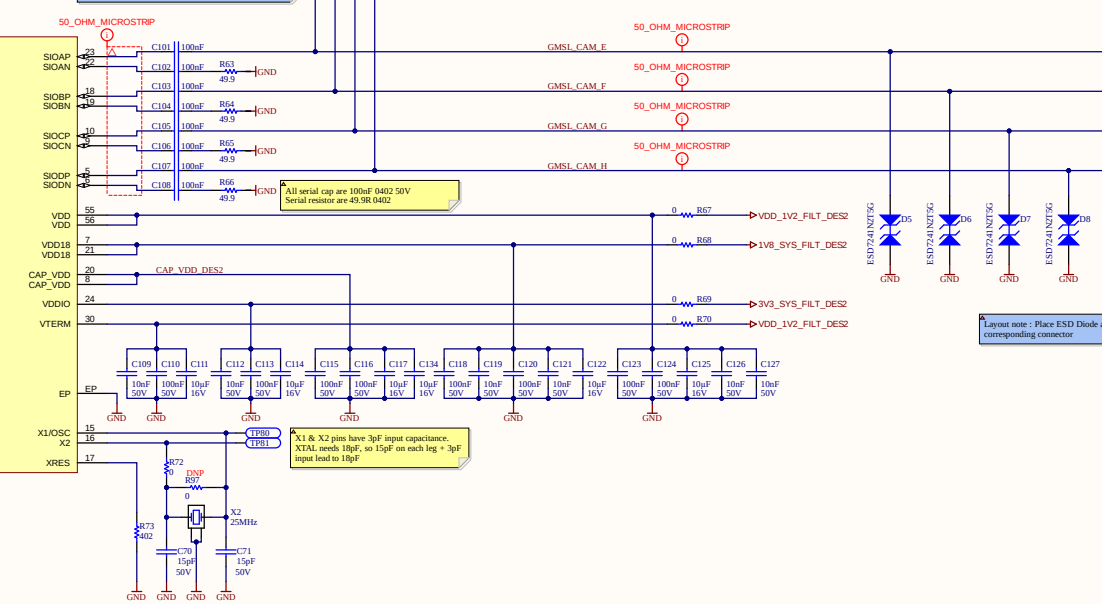


12V INJECTION (PoC)



PoC filter design from MAXIMHW design guide page 37 for 12V 1A
<https://www.analog.com/media/en/technical-documentation/user-guides/public-gmsl2-hardware-design-and-validation-guide.pdf>

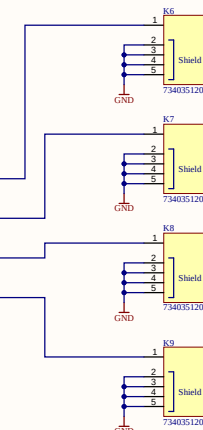
Layout note : Place link capacitor as close as possible to its corresponding SIO Pin.



All serial cap are 100nF 0402 50V
Serial resistor are 40.3R 0402

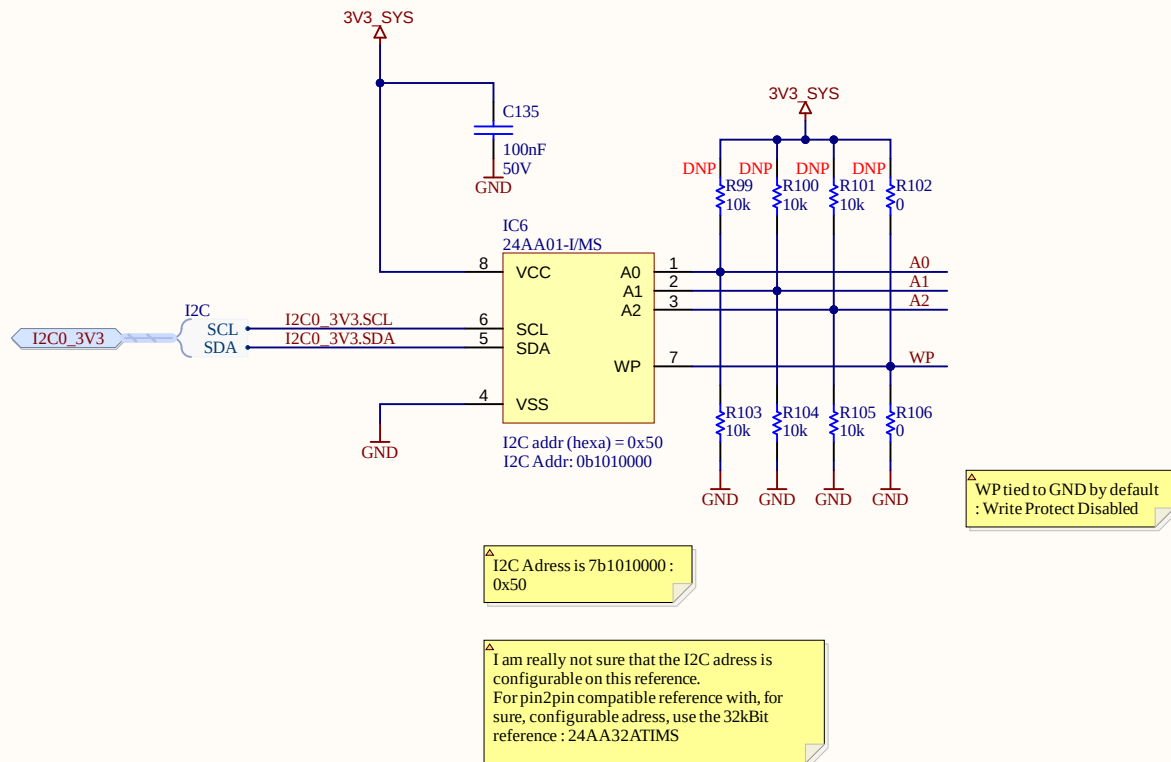
X1 & X2 pins have 3pF input capacitance.
X1AL needs 10pF, so 15pF on each leg + 3pF input lead to 10pF

Layout note : Place ESD Diode as close as possible to its corresponding connector



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